

LSFP-Tracker: An Autonomous Laser Stripe Feature Point Extraction Algorithm Based on Siamese Network for Robotic Welding Seam Tracking

Runquan Xiao , Yanling Xu , Fengjing Xu, Zhen Hou, Huajun Zhang, and Shanben Chen 

Abstract—Laser vision-based welding seam tracking has become the key technology to realize intelligent robotic welding, and extracting the feature point of welding seam is an essential step. Aiming at the low flexibility of classical methods, this article proposed a laser stripe feature point tracker (LSFP-tracker) based on siamese network for robotic welding seam tracking. First, a lightweight feature extraction network is constructed to produce high resolution feature maps that contain both low-level geometric details and high-level semantic information, which can ensure model's tracking accuracy and antinoise capability. Moreover, a feature refinement module is designed, which can enhance the tracking stability on specific seams through channel-wise emphasizing useful features. Subsequently, stripe's curvature and three preset templates were introduced to adaptively detect the initial feature point and corresponding bounding box, which is helpful to enlarge the applicability scope of LSFP-tracker. The experimental results show that the LSFP-tracker can maintain satisfying tracking performance on various welding seams, such as butt, V-groove, multi-pass seams, etc., even under the interference of welding noise. And the comparison tests prove the advantage of LSFP-tracker, which has great application potential in practical welding.

Index Terms—Feature extraction, laser visual sensor, robotic welding, seam tracking, siamese network.

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I. INTRODUCTION

WITH the development of modern manufacturing, intelligent robotic welding has become the goal pursued by scholars around the world. Visual sensor-based welding guidance, welding seam tracking, and welding pool monitoring are becoming the key technologies to realize intelligent welding [1]. As the laser vision sensor (LVS) based seam tracking technology has the advantages of simple control structure and high precision [2], it has become the most promising choice in practical welding [3].

Extracting the feature point of welding seam from laser stripe images is one of the essential steps in LVS-based seam tracking [4]. A practical solution is to extract and fit the feature points based on the geometrical features of the stripe. Wu et al. [5] introduced a modified Hough algorithm to detect the feature points. Muciz [6] extracted the feature points of narrow butt seam based on the gray value's difference. Moreover, to enhance the tracking stability, Chen et al. [7] carefully designed a convolution kernel to search the laser stripe and feature point of fillet welding seam, which can cope with the strong welding noise interruption. Generally, these extractors are properly designed for specific seam types, which lacks the flexibility for different seam types. Then, to extend the application range of the algorithm, Li et al. [8] proposed a similarity measurement method to filter welding noise, and defined the corresponding string types to identify typical welding seams. Wang et al. [9] extracted the laser stripe based on LoG filter, and then three typical welding seams were summarized for feature point detection. Our previous work [10] designed an improved Snake model to detect laser stripes, and the curvature was introduced for feature point extraction. These extractors are based on manually designed features, while the features produced by the convolutional neural networks (ConvNet) have shown their effectiveness in welding noise reduction. Wu et al. [11] designed a laser stripe edge guidance network to segment stripes. Zhao et al. [12] trained an ERFNet to extract the stripe and feature points simultaneously.

In recent years, a variety of target tracking algorithms have been introduced to track feature points, which presented satisfying tracking performances even under welding noise disturbance. In these studies, the feature points were first extracted in the initial frame, and the bounding boxes (Bbox) were generated

TABLE I
SUMMARY OF MAIN FEATURE EXTRACTION METHODS

Extractor based		Tracker based	
Key techniques	Test joints	Key techniques	Test joints
String matching [8]	V-groove	KCF [13]	Butt & Lap
LoG filter [9]	Fillet joints	DCFNet [17]	Multi-pass seams
Snake model [10]	Multi-pass seams	STC [19]	Intersecting seam
ERFNet [12]	Multi-pass seams	ECO [21]	Butt, Lap & Fillet

accordingly. Then, the variations of the feature points were searched in subsequent images based on the similarity of laser stripes. Zou et al. [13] used the kernelized correlation filter (KCF) [14] to track lap welding seam, and used fuzzy control to promote the tracking stability. In his later research [15], a more powerful efficient convolution operator (ECO) [16] that embedded C-GAN for welding noise suppression was proposed, which increased the overall tracking accuracy. Similarly, Xu et al. [17] introduced generative adversarial network (GAN) to DCFNet [18] to track feature points of multi-pass welding seams, which achieves satisfying accuracy even under the interference of welding noise. Compared with these GAN-based denoise methods, the state and measurement equations are more convenient approaches to enhance the tracking performance. Jia et al. [19] realized intersecting seam tracking with extending and adopting kalman filter (EAKF) boosted spatio-temporal context (STC) [20] tracker. Fan et al. [21] verified the outstanding applicability of ECO-tracker improved by particle filter (PF) in various weld types.

Generally, these extractor based or tracker based feature point methods, as given in Table I, are optional solutions for tracking specific welding seams. Considering the wide variety of seam types, it is difficult to specifically design extraction algorithms for each seam. Contrarily, tracker-based frameworks utilize the similarity of stripes for tracking, which would not be limited to seam types and has broad application prospects.

However, above mentioned tracker-based methods still have their limitations. On the one hand, the employed trackers, such as KCF, STC, and ECO, used specific features, such as histogram of oriented gradient (HOG) or CN (Color Name), to find the most similar position through correlation filtering. Since these features are insufficient to distinguish the stripe from welding noise, they are not particularly suitable for stripe feature point tracking tasks. As a result, these approaches relied on techniques, such as GAN and PF, to overcome the welding noise's interference. On the other hand, the generated initial Bboxes are based on specifically designed extraction methods, which limits the applicability scope of the entire framework.

Aiming at these problems, this article proposed an adaptive tracking framework based on siamese tracking framework [22]. The main contributions of this method consist of two aspects. First, a laser stripe feature point tracker (LSFP-tracker) was proposed. A lightweight feature extraction network for laser stripe images (LSFENet) is properly constructed to produce features suitable for seam tracking while maintaining efficiency. And an online feature refinement module (FRM) is present to improve the tracking performance. Second, an adaptive initial Bboxes generation method based on curvature and LSFENet is

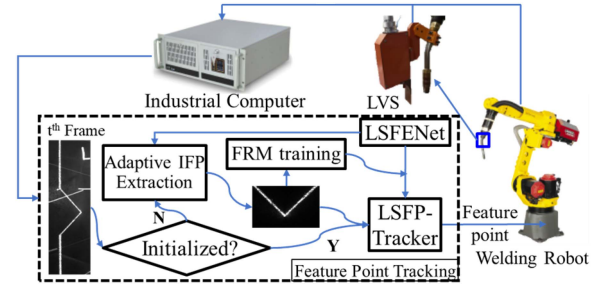


Fig. 1. Diagram of LSFP-tracker-based welding seam tracking system.

designed, which extends the application range of the tracking framework to various welding seams.

The rest of this article is organized as follows. Section II introduces the LVS-based seam tracking system. In Section III, the properly designed LSFP-tracker is discussed in detail. Then, the adaptive initial feature point (IFP) extraction algorithm is designed in Section IV. Several experiments are conducted to evaluate the performance and advantage of the proposed method in Section V. Finally, Section VI concludes this article.

II. LASER VISION-BASED SEAM TRACKING SYSTEM

The experimental system mainly consists of three parts: an industrial personal computer (IPC); a LVS; and a welding robot, as presented in Fig. 1. The vision sensor is rigidly mounted on the robotic manipulator, forming an eye-in-hand system. The welding robot, working as the motion actuator, will correct the welding trajectory according to the coordinates received from IPC.

The LVS is composed of an industrial charge coupled device (CCD) camera and a line laser. A narrow band optical filter is installed to reduce arc interference during welding. The vision sensor's measurement model is as (1), which converts a laser stripe point's pixel coordinate $p_I = [x, y, 1]^T$ to three-dimensional (3-D) coordinate $P_b = [X_b, Y_b, Z_b, 1]^T$ in the robot base coordinate system (S_b)

$$P_b = {}^bT_t {}^tT_I p_I \quad (1)$$

where, bT_t is the conversion matrix from the tool coordinate system (S_t) to S_b , which depends on robotic posture. tT_I denotes the homography that maps p_I to its 3-D coordinate in S_t . The measurement model is calibrated based on the method described in our previous works [23].

The IPC is the core of the system. During seam tracking, a program running on IPC will extract the feature points from the laser stripe images captured by the vision sensor. The acquired points are then transferred to S_b . Finally, the 3-D points are stored in a first in first out (FIFO) sequence, and then fed back to the robot to form a closed-loop motion control.

The feature point extraction procedure mainly consists of two steps: In the first frame, the IFP and corresponding Bbox will be adaptively extracted, and the tracking bounding box is generated accordingly. LSFP-tracker is then applied to detect the feature points in subsequent images.

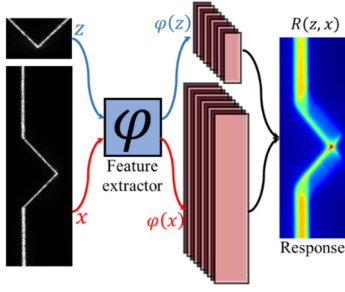


Fig. 2. Architecture of SiamFC.

III. LASER STRIPE FEATURE POINT TRACKER

As mentioned earlier, due to the wide range of weld types, it is impossible to design a specific feature point extractor for each welding seam. Therefore, it is more suitable to treat it as a target tracking problem. In this article, we based on the siamese tracking framework (SiamFC), and put forward the LSFP-tracker.

A. Brief Review of SiamFC

SiamFC is a framework proposed by Bertinetto et al. [22] that uses a siamese network structure as the similarity comparison function to predict the position most similar to the exemplar target on the candidate search image.

The basic architecture of SiamFC is shown in Fig. 2, it takes a pair of images as input, comprising a target Z and a search image X . And a trained ConvNet φ is used to extract the features of the two inputs, denoted as $\varphi(Z)$ and $\varphi(X)$, respectively. Finally, $\varphi(Z)$ is used as the kernel to convolve the searching feature map $\varphi(X)$, defined as

$$R(Z, X) = \varphi(Z) * \varphi(X) \quad (2)$$

where $*$ denotes the convolution operation. $R(Z, X)$ is the output response map that describes the similarity between Z and X . The higher the response score, the more likely the sub-patch is similar to Z .

Based on convolution operation, SiamFC has the advantages of simple structure and high computational efficiency. Moreover, thanks to its end-to-end architecture, it is more convenient for SiamFC to adjust convolutional neural network (CNN) features compared with the correlation filter-based trackers, such as KCF and STC. During tracking, only $\varphi(X)$ need to be continuously calculated, which further accelerates the inference speed of the model and ensures it meets real-time requirements.

In this article, the feature extractor of SiamFC is properly redesigned for laser stripe tracking tasks, and a FRM is embedded for online feature adjustment.

B. Laser Stripe Feature Extraction Network

In traditional SiamFC, the feature extractor is a pre-trained ConvNet, such as VGGNet, AlexNet, etc. However, as these ConvNets are designed for classification tasks, the extracted features may perform poorly when tracking stripe images.

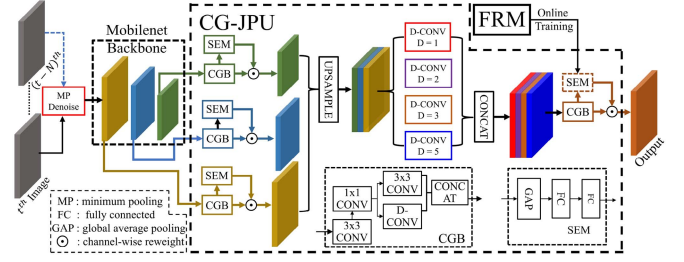


Fig. 3. Structure of laser stripe feature extraction network.

On the one hand, these ConvNets are mostly trained on classification datasets, which makes the output features not fully suitable for tracking tasks. The extractor will focus on the differences between categories, while the features required for tracking tasks should also provide some discriminative capability to similar objects.

On the other hand, the structure of traditional ConvNet aim to generate high-level features has semantic information with strong representation capacity. But their feature maps have low resolution, which not only limits the template size, but also reduces the tracking accuracy. In (2) the target position is searched in $\varphi(X)$. If the resolution of $\varphi(X)$ is $1/2^n$ the size of X , theoretically, there will be an uncertainty of n pixels in the predicted position.

Normally, the low-level features of ConvNet with high resolution contain more geometric details that are beneficial to improve tracking accuracy, while high-level features with large receptive field, i.e., the region of input image that affects the output features, are robust to the noise. Therefore, with the consideration of inference speed, the LSFENet was designed as the feature extraction network for laser stripe images. As shown in Fig. 3 it mainly includes three parts: denoise module, feature extraction backbone, and feature fusion module.

1) *Denoise Module*: As welding noises are similar to laser stripes, and LSFENet's structure is simple, these noises are unlikely to be fully suppressed. Commonly, welding noise only exists for a short time, while the laser stripes are relatively stable between adjacent frames. Therefore, a channel-wise minimum pooling is proposed as the denoise module, as

$${}^t F_{\text{out}} = \min \{{}^{t-N} F_{\text{in}}, \dots, {}^t F_{\text{in}}\} \quad (3)$$

where ${}^t F_{\text{in}}$ and ${}^t F_{\text{out}}$ are the t^{th} input and output feature map respectively; N is the number of adjacent frames; and $c \in \{1, 2, 3\}$ is the channel number.

This module treats the welding noises and stripes as moving foreground and background respectively, and uses their difference in time sequence to filter the welding noise. Through the denoise module, the welding noise can be effectively suppressed without introducing additional computational costs.

2) *Feature Extraction Backbone*: To ensure the efficiency of the network and the tracking accuracy, we use the output features of the first, third, and fifth layers of MobileNet [24] as the backbone. The resolution of these feature

maps will become 1x, 0.5x, and 0.25x of the input size accordingly.

As mentioned earlier, these low-level features are effective for target localization. However, these features have small receptive fields, which are unable to differentiate the target from the welding noise. Therefore, we design a feature fusion module context guided joint pyramid upsampling (CG-JPU) to enhance model's discriminative ability.

3) *Feature Fusion Module CG-JPU*: The three backbone feature maps are first preprocessed by a context guided block (CGB) [25], and then up-sampled to the input size and stacked together to obtain a high-resolution map that contains multiscale features. Subsequently, dilated convolutions (D-Conv) with different dilation rates are applied to produce multi-scale semantic information.

Theoretically, the receptive field Rf_l of l^{th} regular convolutional (Conv) layer can be calculated as:

$$Rf_l = Rf_{l-1} + (k_l - 1) * \prod_{i=1}^{l-1} s_i \quad (4)$$

where k_l is the kernel size and s_i is the stride step.

As for a D-Conv layer with dilation rate α , its equivalent kernel size k'_l is equal to

$$k'_l = \alpha (k_l - 1) + 1. \quad (5)$$

Therefore, (4) can be rewritten as

$$Rf_l = Rf_{l-1} + \alpha (k_l - 1) * \prod_{i=1}^{l-1} s_i. \quad (6)$$

It can be seen that D-Conv can significantly enlarge receptive fields without introducing additional parameters. And to avoid the gridding effect, the dilation rates are set as 1, 2, 3, and 5, respectively. Finally, the four D-Convs' outputs are stacked together, and fused by a Conv to produce output feature maps. The obtained feature maps contain multi-scale context information, which leads to better performance [26].

The introduced CGB is helpful to preserve texture information passing through the model. The CGB stacks the outputs of Conv and D-Conv to learn local and surrounding texture features simultaneously. Then, a squeeze-and-excitation module (SEM) [27] is used to recalibrate the output features, defined as

$$\begin{aligned} F_{out} &= \mathbb{W} F_{in} \\ &= \mathcal{S}_g \left(\mathcal{W}_2 \mathcal{S}_g \left(\mathcal{W}_1 \sum_{x=1}^w \sum_{y=1}^h F_{in}(x, y) / (wh) \right) \right) F_{in} \end{aligned} \quad (7)$$

where $\mathbb{W}$ is the calculated weight of each channel. $\mathcal{W}_1$ and $\mathcal{W}_2$ are the parameters of SEM. w and h are the wide and height of F_{in} , respectively. During training, SEM will learn to balance the local and surrounding features. Through channel-wise emphasizing useful components, the discriminative ability of re-weighted features to stripe and noises will be enhanced.

Moreover, according to (6), the D-Conv used in each layer will exponentially expand the model's receptive fields, while

maintaining the resolution and computational efficiency. Eventually, the obtained feature map contains low-level geometric details and high-level semantic information, which helps maintain tracking performance when facing welding noise disturbance.

Summarily, the size of LSFENet is less than 0.6 MB, and for a 1000*300 image, one forward time is less than 8ms. The lightweight design ensures that LSFP-tracker can meet real-time requirements of the weld tracking process.

C. Online Feature Refinement Module

LSFENet will be trained with a dataset containing various laser stripe images to ensure its adaptability to multiple welding seams. Therefore, each feature map describes a particular stripe pattern and will contribute differently when tracking a specific seam. Additionally, the convolution kernel $\varphi(\mathcal{Z})$ is generated from the features surrounding the target, which ignores the information around the Bbox. This makes the obtained $\varphi(\mathcal{Z})$ similar to a classifier trained only with positive samples, which reduces its discriminative ability for cluttered backgrounds.

Therefore, by properly increasing the contribution of target-related features, and enlarging the distance between the target and surrounding features, the model can focus on the discriminative features, which is helpful to improve the tracking performance. To this end, the last SEM, as shown in brown dotted line in Fig. 3, is used to modulate the output features.

It is expected that features closer to the target points will have higher response values. Therefore, a Gaussian map is generated at the target, defined in (8).

$$Y_{(x,y)} = \exp \left(\left((x - x_0)^2 + (y - y_0)^2 \right) / (2\sigma^2) \right) \quad (8)$$

where σ is the standard deviation, which is related to the size of the Bbox. (x_0, y_0) is the position of the target point.

Then, the SEM is finetuned by forcing the output response map close to the Gaussian label map. The loss function for this training process is defined as

$$L_{reg} = \|Y - \mathbb{W}_Z \varphi(\mathcal{Z}) * \mathbb{W}_X \varphi(\mathcal{X})\| \quad (9)$$

where $\mathbb{W}_Z$ and $\mathbb{W}_X$ are the weights outputted by SEM. And $\mathbb{W}_Z \varphi(\mathcal{Z}) * \mathbb{W}_X \varphi(\mathcal{X})$ is the refined response map.

As the model trains, (9) will force the remodulated response map approaching Y . Then, the target-related features can be effectively enhanced, while the similarity between the target and background can be minimized, which can enhance the discriminative property of $\varphi(\mathcal{Z})$.

It is worth mentioning that the above training process is only performed once after determining the tracking target. The weight $\mathbb{W}$ of each feature's channel is obtained by SEM that performs nonlinear transformation to the input feature map. Since the parameters of SEM are the optimization goals of L_{reg} , SEM can also learn the mapping between the input features and their relevance to the target when minimizing L_{reg} . This ensures the fine-tuned SEM has the ability to automatically emphasize target-related features in subsequent images.

As shown in Fig. 4, the re-calibrated features are able to separate target from similarities. We randomly selected 20 laser stripe images of fillet welding seam, and used t-SNE [23] to

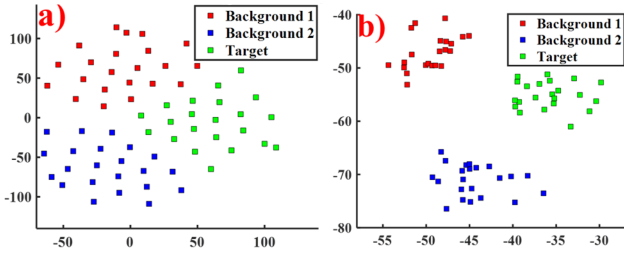


Fig. 4. Output features (a) without and (b) with FRM were visualized using t-SNE. Blue dots represent targets' features, and the red and blue dots represent similarities' features.

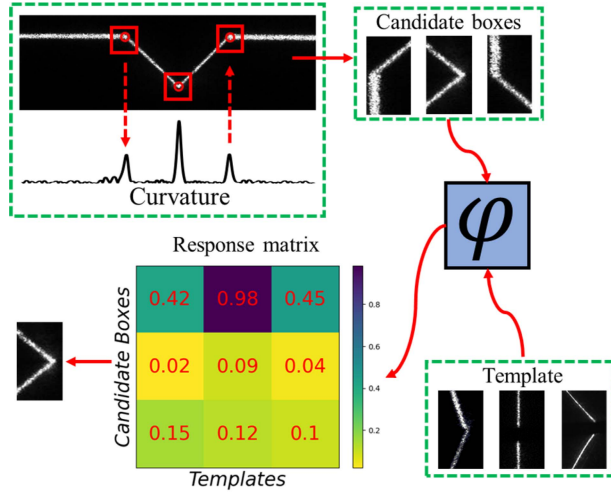


Fig. 5. IFP and corresponding Bbox detection based on LSFENet.

visualize the features of the target and its surrounding stripes. Compared with the directly extracted features, the recalibrated features of the same category are more centralized, and the distance between different clusters gets larger. This reduces the interference of similar stripes on the tracking results, which is conducive to improving tracking stability.

IV. ADAPTIVE INITIAL BOUNDING BOX EXTRACTION

LSFP-tracker needs to determine an initial bounding box containing the tracking target in the first frame. This section will discuss an adaptive IFP and corresponding Bbox extraction method suitable for various welding seams, as shown in Fig. 5. The candidate feature points are first extracted based on the geometric features. Then, the candidate Bboxes are matched with preset templates to determine the tracking target.

A. Candidate Feature Points Extraction

Since the laser stripe images before welding are less polluted by welding noises, the traditional geometric feature-based method is able to extract the laser stripes.

Considering that the profiles of laser stripes are close to Gaussian distribution, the center of the stripe is the zero point of gray value's first derivative along the normal direction [29]. The normal direction, denoted as $[n_x, n_y]^T$, can be determined by

using the Hessian matrix H of the image. And H can be derived by convolving the image I with Gaussian partial derivative kernels $\mathcal{G}$, as

$$H = \begin{bmatrix} d_{xx} & d_{xy} \\ d_{xy} & d_{yy} \end{bmatrix}, L = \mathcal{G} * I \quad (10)$$

where d_{xx} and d_{yy} are the 2nd-order partial derivatives of I along the x and y directions, respectively, and d_{xy} is the mixed partial second-order derivatives.

Then, the normal direction equals to the eigenvector corresponding to the eigenvalue of maximum absolute value of H . Hence, the zero point of gray value's first derivative along $[n_x, n_y]^T$ can be calculated as

$$p = -(d_x n_x + d_y n_y) / (d_{xx} n_x^2 + 2d_{xy} n_x n_y + d_{yy} n_y^2) \quad (11)$$

where d_x and d_y are the first-order partial derivatives of I in the x - and y -directions, respectively. If $|p| < 0.5$, it can be declared as the stripe's center point.

Finally, (12) is used to record all possible center points by traversing the image from the second row to the last row. Then, backtracking from the minimum position of the last row in M , the laser stripe can be obtained based on seam carving [30]

$$M_{x,y} = C(x, y) + \min(M_{x-1,y-1}, M_{x-1,y}, M_{x-1,y+1}) \quad (12)$$

where $C(x, y) = -1$ if $|p| < 0.5$, otherwise 0.

Subsequently, the candidate feature points are extracted based on the curvature distribution of the stripes, calculated as

$$\kappa = |s''| / (1 + s'^2)^{1.5} \quad (13)$$

where s' and s'' are the first and second order derivatives of the extracted stripe.

The feature points are generally the inflections discontinuous points of the stripes. According to the geometric relationship, these points are the local maxima of the curvature. The larger the peak of the curvature is, the more dramatically the curve changes, and the more likely it is to be the feature point. Therefore, the points whose curvatures are bigger than a certain threshold are reserved as candidate feature points, as shown in the left-dotted box in Fig. 5.

B. Initial Bounding Box Detection

The candidate bounding boxes are generated at the candidate feature points with aspect ratio of 3:1, specifically defined as

$$\text{Rect}(x, y) = \{x - 3\omega/2, y - \omega/4, 3\omega, \omega\} \quad (14)$$

where ω is the width of the bounding box.

Since the stripes are slender from top to bottom across the image, Bbox in this ratio allows it to contain more stripe features without increasing its area. Moreover, the feature point is set at 1/4 width, which is also helpful to improve the density of stripe in Bbox. Consequently, the obtained convolution kernel $\varphi(\mathcal{Z})$ will contain sufficient features, so as to improve the tracking efficiency without reducing its anti-noise capability.

Then, the features of the candidate Bboxes are extracted by LSFENet and compared to three preset templates. Then, the

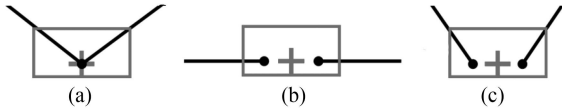


Fig. 6. Three preset templates and corresponding feature points.

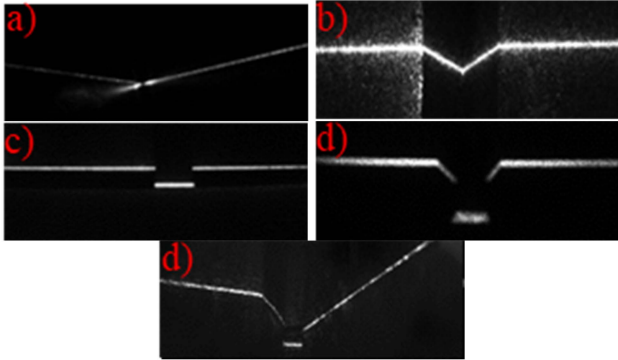


Fig. 7. Laser stripe images of (a) fillet, (b) V-groove, (c) butt, (d) gap V-groove, and (e) half-sided V-groove welding seams.

tracking target can be determined by the maximum response pair of the candidate Bbox and template.

After observing the geometry of feature points on different welding seams, three templates are summarized in Fig. 6 for matching. Among them, the feature point of Template-A is the candidate point with the maximum response. And the feature points of template-B and template-C are the center of two maximum response points.

These templates are suitable for most seam types: template-A is suitable for the majority of continuous stripes, such as V-groove and fillet seam [see Fig. 7(a) and (b)]. Template-B is responsible for butt seam [see Fig. 7(c)]. Template-C can be used to match gap seams, such as gap V-groove and half-sided V-groove seam [see Fig. 7(d) and (e)].

It is worth mentioning that the method described above has higher accuracy compared with using the preset templates to detect IFP directly. Considering the different morphology of grooves, the shape of preset templates cannot fully match the stripe to be detected. It is difficult for the templates to accurately react to the changes of groove geometrical characteristics, such as groove angle or thickness, which will lead to the change of feature point's location. In these cases, the template may not have a maximum response at the feature point in search image. Since LSFNet uses a specific position in the Bbox as the feature point, these matching errors will become systematic errors that persist in the tracking process.

V. EXPERIMENTAL VERIFICATION AND ANALYSIS

A. Experimental System

To verify the practicability of the proposed tracking algorithm, a laser vision-based seam tracking system was developed, as shown in Fig. 8. The Fanuc robot equipped with the Fronius CMT welding power is the welding actuator. The IPC has 8G

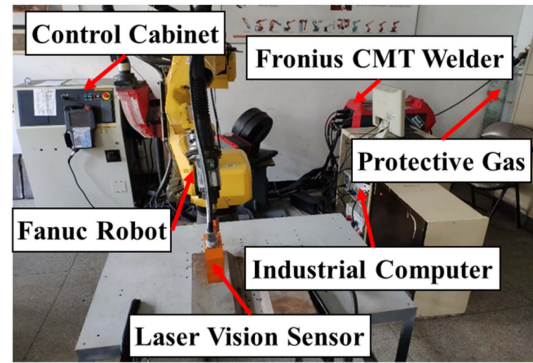


Fig. 8. Laser vision-based seam tracking system.

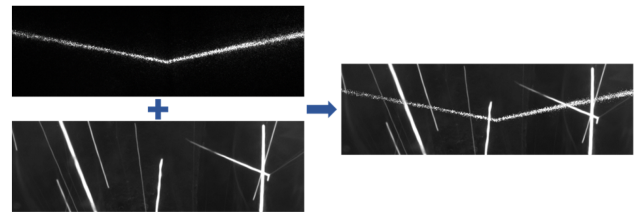


Fig. 9. Schematic of synthetic welding laser stripe image.

memory and an Intel Core I7-6700 processor. And LSFNet runs on GTX 1050Ti GPU with 4G memory. The IPC is connected with the robot through Robot Interface to read the real-time posture of the robot and write position registers at a frequency of 80 Hz. The visual sensor is rigidly installed in front of the welding torch, and its working distance is around 80 mm. The sampling rate is 40 fps with an image size of 1236*1628. To speed up the efficiency, an ROI of size 400*1000 is chosen.

B. Training of LSFNet

- Dataset Preparation:** To ensure the trained model's performance, a dataset containing various types of laser stripe images was prepared. Despite the images captured during actual welding processes, the dataset also includes the synthetic images built to reduce welding tests, as shown in Fig. 9. Before welding, the laser stripe images were collected by scanning the workpiece. During welding, the laser power is shut off, and the images only containing welding noises are acquired. Finally, the noise images are overlaid on the stripe images to simulate the images in actual welding process. Besides, data augmentation techniques including random flipping, adjusting contrast, and adding Gaussian noise were utilized to improve the trained model's generalization. Finally, the dataset is split into train and test sets in a ratio of 6:4.
- Training Strategy and Results:** To improve training efficiency and ensure the performance of LSFNet, we train the feature extraction backbone and CG-JPU module separately, and the training parameters are given in Table II. First, MobileNet is finetuned via transfer learning on a classification task. The images containing both laser stripes and welding

TABLE II
TRAINING PARAMETER SETTINGS

Parameters	MobileNet	CG-JPU	FRM
Batch size	8	4	1
Max epochs	100	300	100
Dropout keep probability	0.95	0.8	1
The initial learning rate	0.0001	0.005	0.01
The momentum value	0.98	0.95	0.9

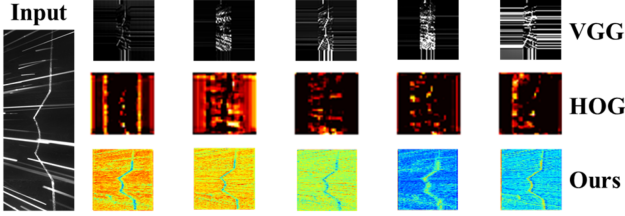


Fig. 10. Feature of HOG, VGG features, and LSFENet.

noises are treated as positive samples, while the images only containing welding noises are negative samples. This training process aims to improve MobileNet's sensitivity to laser stripes. Subsequently, the parameters of the feature extraction backbone are frozen, and a Conv is added behind CG-JPU. Then, by taking the synthetic images as input and the corresponding laser stripe images as labels to train the model on the stripe segmentation task. During training, the learned texture features of CG-JPU will be able to differentiate the stripes from the welding noise. As shown in Fig. 10, the features extracted by LSFENet can preserve more textural information compared with HOG or visual geometry group (VGG) features. The laser stripe will gain more response than the welding noises.

C. Algorithm Performance Verification

The feature point tracking experiment was first conducted to verify the accuracy of LSFP-tracker. A sequence of laser stripe images was recorded during welding, and the feature points on these images were labeled manually as reference. The Euclidean distance between the extracted point $\hat{p}$ and the reference p was treated as extraction error, as

$$E(\hat{p}, p) = \|\hat{p} - p\|_2. \quad (15)$$

The results are shown in Fig. 11, it can be seen that the extraction errors are within 3 pixels under the interference of welding noise. Since the resolution of LVS is around 0.04 mm/pixel at 80 mm working distance, the actual error is less than 0.12 mm. Moreover, the mean and variance of the tracking error are rather low (about 1.1 and 0.4 pixels, respectively), which proved that LSFP-tracker can retain satisfactory tracking stability, even with the interference of welding noises, such as the tenth sample frame.

Subsequently, the adaptability and robustness of LSFP-tracker were tested on six kinds of laser stripe images, including four typical welding seams, and the second and third passes of multi-pass welding (MP2-1 and MP2-2, respectively), as given in Table III. The red Bboxes are generated by the method

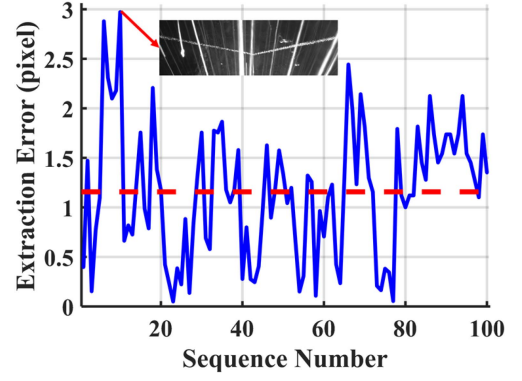


Fig. 11. Feature points tracking accuracy on real welding image sequence.

TABLE III
TRACKING RESULTS ON DIFFERENT WELDING SEAMS

Types	Laser stripe images	MTE
V-groove		0.26
Fillet		0.25
Butt		0.40
Gap V-groove		0.34
MP2-1		0.27
MP2-2		0.26

described in Section IV. And the FRM is trained with the parameters given in Table II. Similar to the procedure of generating a synthetic dataset, a laser stripe image, whose feature point was manually marked, was used as the source, and different welding noises were overlaid on it. Besides, the stripes are randomly shifted between 0-10pixels to imitate the change of feature points during welding. The tracking performance was evaluated by mean tracking error(MTE), as (16). The results are given in Table III, and the tracking results on fillet and V-groove joints are displayed in Fig. 12

$$\text{MTE} = \frac{1}{n} \sum_{i=1}^n E(\hat{p}_i, p_i) \quad (16)$$

where n is the number of images in the dataset.

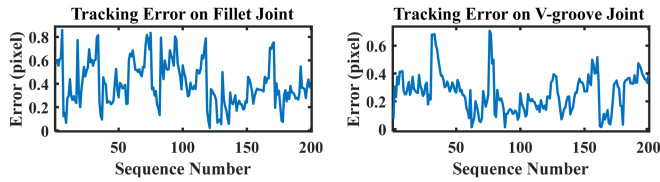


Fig. 12. Tracking accuracy on synthetic fillet and V-groove dataset with different welding noise interference.

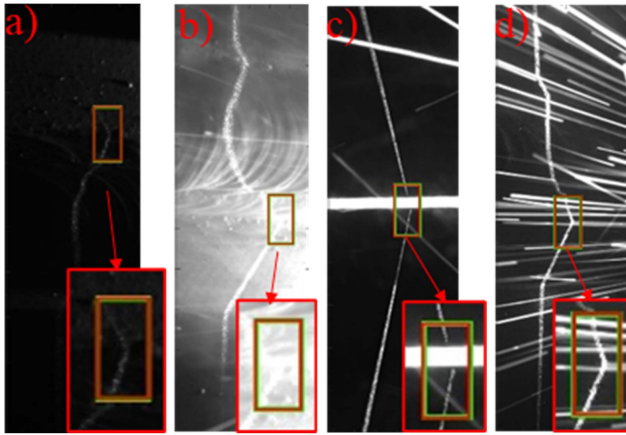


Fig. 13. Feature point tracking results in special cases. The red and green Bboxes are labels and tracking results, respectively.

It can be seen that LSFP-tracker is suitable for different welding seams, and can maintain satisfying accuracy. Since LSFP-tracker is based on the similarity of laser stripes, it is not sensitive to certain seam types. The error can be kept within 1 pixel, which demonstrates the exceptional resistance of LSFP-tracker to welding noise. As the locations of feature points are known in the synthetic dataset, the tracking error is slightly lower than that in real welding process.

It is worth mentioning that it's no need to adjust any super-parameters when tracking different welding seams, which further proves the adaptability of LSFP-tracker.

Besides, LSFP-tracker will still be effective in some special cases. As shown in Fig. 13(a), due to the short exposure time, the image is in low intensity, and the stripe becomes nearly unseen. Since LSFENet is trained on the segmentation task, it will automatically extract the stripe-relevant features. Meanwhile, when some stripes are partially ruined by welding noise, as shown in Fig. 13(b)–(d), LSFP-tracker can still locate the feature points accurately. As all features inside the Bbox are utilized to track the feature point, LSFP-tracker is ensured to cope with local features occlusion.

Finally, tracking experiments were carried out to quantitatively verify the accuracy of LSFP-tracker. A total of 200 sampling points were taken on a V-groove welding seam. The 3-D coordinates of these points are extracted and calculated with (1). The actual coordinates were recorded through teaching. The Euclidean distances between the calculated and actual values are treated as tracking errors. Moreover, after recording the actual welding path, deviation was added to the teaching trajectory.

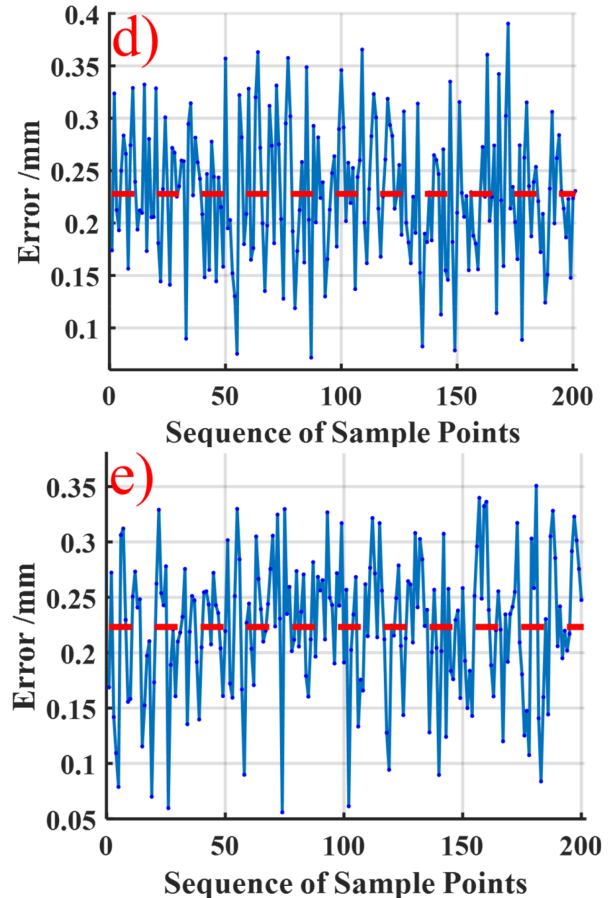
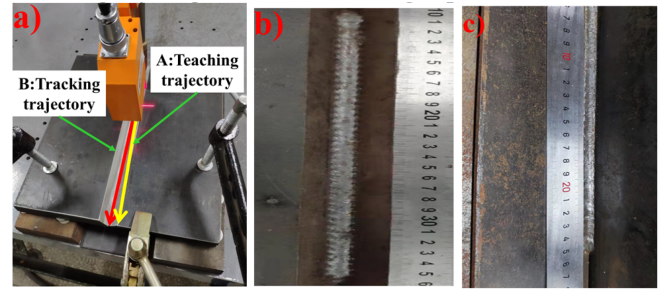


Fig. 14. Tracking experiments. (a) Experimental setup. (b) and (c) Tracking result and accuracy of V-groove seam. (d) and (e) Tracking result and accuracy of fillet seam.

During welding, the visual sensor was used to correct the weld trajectory, as illustrated in Fig. 14(a).

The results in Fig. 14(b) and (c) show that the average tracking error is 0.23 mm. Similarly, the MTE in fillet seam is 0.22 mm, as shown in Fig. 16(d) and (e). Considering the calibration error, the error evaluated in real tracking experiment will be slightly larger than that in image space.

D. Ablation Experiments

The ablation experiments were then performed to verify the effectiveness of each module of LSFP-tracker. Similarly, tracking accuracy is evaluated with MTE on the dataset containing six kinds of synthetic laser stripe images.

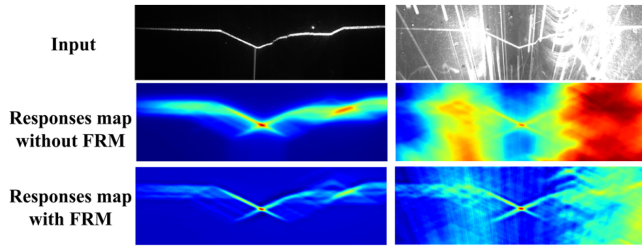


Fig. 15. Response maps with and without the FRM.

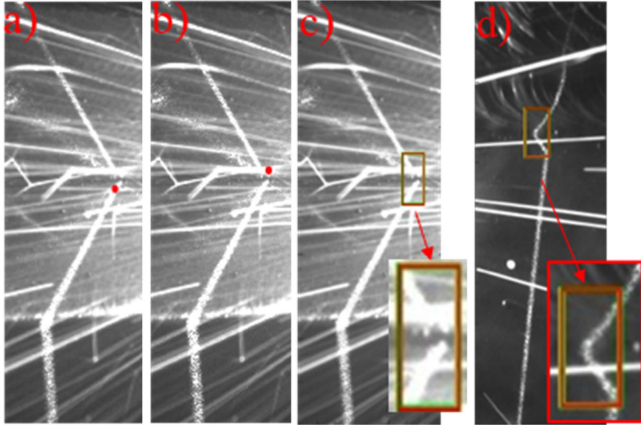


Fig. 16. Red dots in (a) and (b) are the extraction results of A_1 [5] and A_2 [8], respectively. (c) and (d) Tracking results of LSFP-tracker.

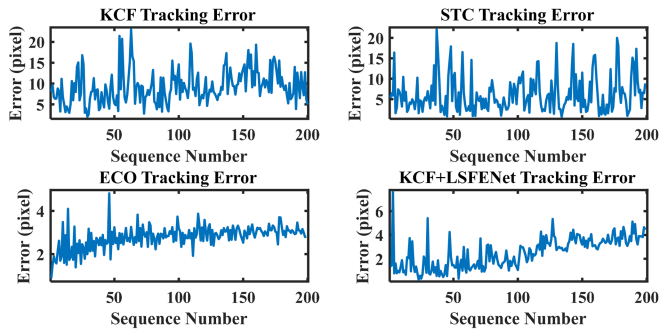


Fig. 17. Tracking results on V-groove welding seam.

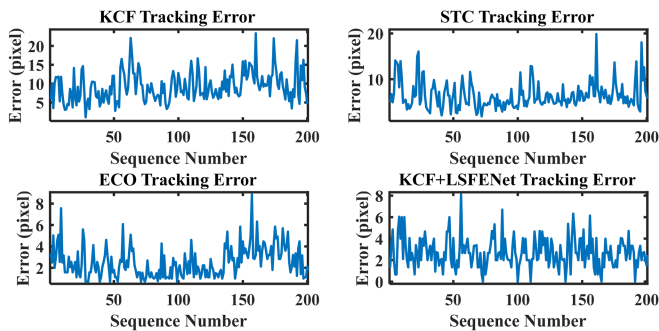


Fig. 18. Tracking results on Fillet welding seam.

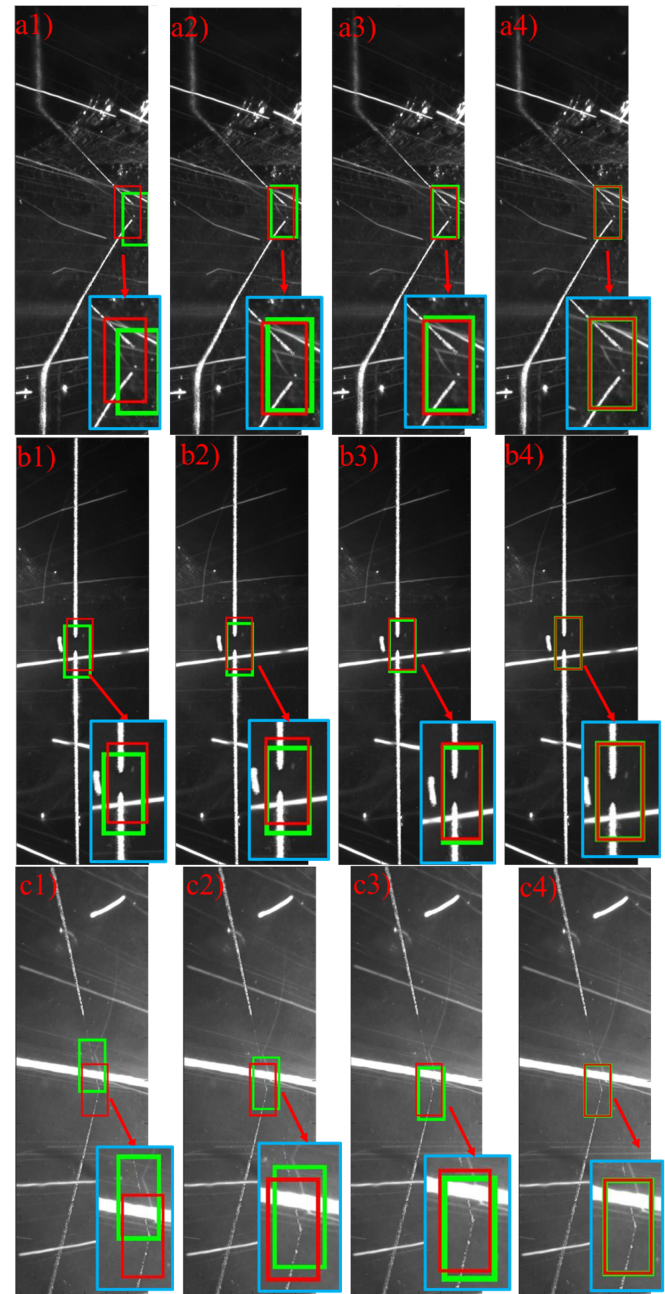


Fig. 19. Tracking results of 1) KCF, 2) STC, 3) ECO, and 4) LSFP-tracker on (a) gap V-groove, (b) butt, and (c) fillet welding seams.

TABLE IV
INFLUENCE OF KERNEL SHAPE ON TRACKING PERFORMANCE

Kernel shape	60*20	75*25	90*30	120*40
Speed (ms)	10.6	18.7	29.1	40.9
MTE (pixel)	1.7	1.1	0.9	0.3

The influence of the kernel shape on tracking efficiency and stability was first tested, and the results were given in Table IV. It can be seen that as kernel size grows larger, the tracking accuracy will increase while the computational efficiency will reduce.

TABLE V
INFLUENCE OF EACH MODULE ON TRACKING PERFORMANCE

Models	M1	M2	M3	M4	M5
Speed (ms)	18.7	18.6	17.9	16.1	25.3
MTE (pixel)	1.1	3.4	7.6	2.3	1.5

Generally, when the size is larger than 75×25 , the extracted features are sufficient to track the feature point.

Then, the influence of each module in LSFP-tracker on the tracking performance is given in Table V. M1 is the original LSFP-tracker. Based on M1, M2 removes the denoise module, M3 deletes the FRM, and M4 and M5 eliminate the CGB modules. Moreover, the kernel shape in M1–M4 is 75×25 , while in M5 it is 90×30 .

The experimental results indicate that the denoise module can enhance the tracking performance of LSFP-tracker, attaching only a tiny amount of computation, which indicates that the minimum pooling is an effective denoise solution in practice. As for the FRM, it can significantly improve the tracking performance. As seen in Fig. 15, this module will modify the distribution of features. In the modulated response maps, the disturbance of strong noises and similar patterns will be suppressed, and the target is more prominent, which enhances the tracking stability. And by comparing M4 and M5, it can be revealed that CGB does not patently improve the tracking performance. Its main contribution is to enlarge model's receptive fields, ensuring that it still obtains decent results when using small tracking kernels.

E. Comparison Experiments

Finally, comparison experiments are conducted to verify the advantages of the proposed algorithm. LSFP-tracker was first compared to traditional geometric feature-based algorithms A_1 [5] and A_2 [8], and the results are shown in Fig. 16.

Traditional algorithms use center-of-gravity method to extract the laser stripe, and fit the feature points based on joints' geometric feature. Since the welding noises are similar to the stripe, these methods may fail in presence of welding noises, as shown in Fig. 16(a) and (b). Comparatively, LSFP-tracker is able to deal with different welding noises and has reliable stability, as shown in Fig. 16(c). Its tracking accuracy and efficiency can meet general welding demands. Moreover, for some special-shaped stripes, as Fig. 16(d), it is difficult to design an appropriate geometry-based extractor. LSFP-tracker can still track the feature points, which indicates its advantages in applicability.

Further, LSFP-tracker was compared to KCF-tracker [13], ECO-tracker [21], and STC-tracker [19] to further verify its advantage. Since these methods still used specific approaches to generate the initial Bboxes, the comparison experiments were performed on the synthetic datasets containing images of fillet, butt, gap V-groove, and V-groove welding seams. The MTE and tracking efficiencies are given in Table VI. Parts of the tracking results are shown in Figs. 17–19.

These tracking methods are based on the similarity of stripes, and have the advantage in adaptability over the geometry-based methods. However, the HOG features used in STC-tracker and KCF-tracker are insufficient to maintain tracking stability. As

TABLE VI
TRACKING PERFORMANCE COMPARISON

Kernel Shape	60*20		120*40	
Performance	MTE (pixel)	Speed (ms)	MTE (pixel)	Speed (ms)
KCF	17.2	5.5	12.5	9.4
STC	9.4	34.1	6.3	124.3
ECO	5.2	9.2	2.7	59.3
KCF + VGGNet	9.1	12.2	6.2	16.1
KCF + LSFENet	4.3	8.3	2.5	14.6
LSFP-tracker	1.7	10.6	0.3	40.9

shown in Fig. 19, they are prone to large errors when facing some welding noises. ECO-tracker-based method has fewer tracking errors. As it combines HOG features and CN features, and introduces a factorized convolution operator to select optimized features, the tracking stability is effectively improved. Comparatively, the proposed LSFP-tracker has good accuracy and stability. Moreover, the proposed IFP extraction method is suitable for many types of welding seams, which further extends the application range of LSFP-tracker. Besides, the lightweight design of LSFENet ensures LSFP-tracker's efficiency, its speed has met the real-time requirement of practical seam tracking.

Further, the features extracted by VGGNet and LSFENet were used to enhance KCF-tracker. Also given in Table VI, the tracking stability has been significantly increased. Compared to VGGNet, the features extracted by LSFENet is more efficient to reduce the tracking error of KCF-tracker, demonstrating LSFENet's suitability for laser stripe tracking tasks.

Basically, the properly designed LSFP-tracker achieves satisfactory tracking performance on different welding seams even under the disturbance of welding noises. The reason that LSFP-tracker has better performance over traditional methods may lie in two parts: on the one hand, the structure and produced features of LSFENet are more suitable for stripe tracking tasks. On the other hand, the FRM will autonomously focus on target-relevant features, and enhance the discriminative capability to target, similarity, and background. Thus, the tracking performance can be further improved. It can fully utilize the advantages of stripe's similarity for feature point tracking, and has great application potential.

VI. CONCLUSION

Aiming at the low flexibility of traditional feature point extraction algorithms in laser vision-based robotic welding seam tracking, an autonomous LSFP-tracker was proposed based on Siamese network. The following conclusions were as follows.

- 1) A feature extraction network (LSFENet) is constructed to produce high-resolution feature map that contains both low-level geometric details and high-level semantic information, which was suitable for stripe tracking tasks. The designed denoise module can effectively boost tracker's anti-noise ability, and the CG-JPU block would significantly enlarge model's receptive fields, which were helpful in maintaining tracking performance.
- 2) An FRM was designed to enhance the tracking stability. When tracking specific seams the target-relevant features was automatically emphasized to increase model's discriminative capability through online training.

- 3) An adaptive IFP and corresponding Bbox extraction method was proposed, which was beneficial to extend the flexibility of LSFP-tracker. The candidate points were first extracted based on stripe's curvature. Three templates were then introduced to locate the feature point of welding seam.
- 4) The experimental results show that LSFP-tracker has achieved good tracking performance in various welding seams even under the interference of strong welding noise. The MTE in six kinds of welding seams, including butt, V-groove, fillet, multipass seams, etc., is 1.1 pixel, while the mean consumed time is 18.7 ms. Comparative experiments prove that LSFP-tracker has the advantages in tracking stability and applicability while satisfying real-time requirements.

Summarily, the proposed LSFP-tracker exerts the advantage of similarity-based tracking, and could be a flexible and reliable solution for fully autonomous feature point tracking in LVS-based seam tracking. In the future, LSFP-tracker's efficiency and stability will be further evaluated and optimized to promote its application in practical welding production.

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